AGENTS FOR IMPACT 2026 · CHALLENGE 1: COMMUNITY CLIMATE RESILIENCE

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When Weather Forecasts Fail: Hyper-Local Climate Resilience

A Decision-Grade AI Assistant for Siting Permanent Municipal Cooling Centers · City of Atlanta

The Municipal Siting Imperative

- **Capital Allocation at Stake:** Atlanta has allocated funding to construct three permanent community cooling resilience centers this budget cycle.
- **The Council Vote in 3 Weeks:** Every district council member demands funding. Siting on political gut feeling invites contentious hearings; siting on raw airport sensor data misallocates millions to unoccupied asphalt.
- **The Microclimate Reality:** Two neighborhoods just 2 miles apart in Atlanta vary by **up to 12°F–15°F** on the exact same summer afternoon due to canopy disparities and impervious surface density.
- **A 20-Year Decision:** Permanent capital projects demand defensible justification that proves physical causality, historical climate exposure, and measurable social equity.



> **Speaker Note (0:00–0:30):** *"Regional forecasts say '102°F tomorrow at the airport.' But planners have capital funds and must decide: which exact intersection, why is it hot, and who cannot escape it? We built a decision-grade agent that survives City Council."*

GOVERNANCE & COMMUNITY REALITY · THE MUNICIPAL GAP

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Surviving The Council Hearing: Beyond Raw Sensors

Bridging the Chasm Between Single-Day Mobile Traverses and Actionable Municipal Policy

The Three Crucial Gaps in Traditional Planning

- **Hazard vs. Exposure vs. Vulnerability:** A 100°F industrial rail yard is physically hot, but lacks vulnerable human residents. A 94°F neighborhood with 35% senior citizens and 40% transit-dependent households represents acute life safety risk.
- **The "Why" Behind Microclimates:** City engineers cannot act on a temperature reading alone. They must know the physical drivers: *Is it 70% impervious surface? Street canyon geometry? Tree canopy deficit?*
- **Displacement Transparency:** When Council asks *"Why wasn't Site 4 funded in District 7?"*, planners must provide defensible, data-backed reasons rather than defensive silence.
- **Ethical Demographics Boundary:** Demographics and race must *never* be used to train predictive thermal models. They belong strictly in the post-hoc equity audit to verify who benefits.



> **Speaker Note (0:30–1:00):** *"If you site cooling centers purely on temperature, you build them in parking lots. If you site them purely on politics, you build where*

AGENTS FOR IMPACT 2026 · INTERDISCIPLINARY TEAM

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Our Team — Specialized Municipal AI Builders

Combining Climate Policy, BigQuery Machine Learning, Cloud Agent Orchestration, and Generative UI

H

Harika

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TEAM LEAD & CLIMATE POLICY

- Framed the municipal decision problem for Atlanta City Council & Dana persona.
- Designed the 3-center capital allocation and 800m pedestrian catchment scope.
- Enforced strict ethical AI governance: keeping demographics out of ML features.
- Authored council-ready proposal requirements and displacement metrics.

FC

Faizan Chundrigar

faizy31796@gmail.com

LEAD ML & DATA SCIENTIST

- Trained BigQuery ML `offset_model_v1` predicting relative ΔT (°F vs city mean).
- Engineered leak-free 5 km Geohash-5 spatial-block cross-validation (MAE: 1.25°F).
- Extracted feature attributions via `ML.EXPLAIN_PREDICT` for physical causality.
- Generated 5,618-point 250m inference grid (`grid_predictions`).

RC

Chintamaneni RC

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PLATFORM & AGENT ARCHITECT

- Architected Google Cloud ADK agent powered by Gemini 2.5 Flash on Cloud Run.
- Built & deployed private, warm MCP Toolbox for Databases (`heat-toolbox`).
- Authored BigQuery Semantic Layer (`dim_tract`, `fact_heat_cell`, `site_hot_days`).
- Implemented zero-trust IAM service accounts & least-privilege DCL schema grants.

NV

Nviradiya

hackathon@googl

FULL-STACK & GENERATIVE UI LEAD

- Built productio Next.js 16 + Re interactive plan dashboard.
- Engineered rea time Voronoi microclimate he map with tract intelligence.
- Constructed B feature attribut waterfall widge equity audit ch
- Deployed `urba` `heat-ui` on Cl Run with live ag streaming telem

Speaker Note (1:00–1:30): *"Our team bridged the entire delivery chain: Harika anchored municipal policy and ethical guardrails; Faizan built the spatial BQML model; RC engineered the ADK agent, MCP Toolbox, and Cloud platform; and Nviradiya brought the executive dashboard to life."*

Cross-Functional Execution: Policy · Data Science · Cloud Infrastructure · Production UI

Repository: github.com/chintamanenirc/A4I2026-challenge-1-walkability

GOOGLE CLOUD NATIVE STACK · PRODUCTION ARCHITECTURE

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Collaborative Municipal AI Architecture

Google Agent Development Kit (ADK) + Gemini 2.5 + BigQuery ML + MCP Toolbox for Databases

🤖 GOOGLE ADK + GEMINI

Orchestrates multi-turn planning, enforces minimum 2.0 km site separation, validates campaign bounds, and drafts municipal policy memos.

⚡ BIGQUERY ML

`ML.EXPLAIN_PREDICT` delivers street-level heat offset predictions and feature-level attributions. Race strictly excluded from training.

🧰 MCP TOOLBOX

Curated, private Cloud Run microservice exposing BigQuery semantic functions (`dim_tract`, `vulnerable_population`, `site_hot_days`).

🗺️ VORONOI SPATIAL UI

Next.js 16 dashboard rendering microclimate heat polygons, live scenario stepper, and interactive comparative modal.

🔒 ZERO-TRUST SECURITY

Workload Identity Federation, Secret Manager API keys, dataset-scoped DCL grants (`heat-agent-sa` & `toolbox-sa`).

🚀 CLOUD RUN PRODUCTION

Warm instances (`min-instances=1`) eliminate cold-starts. Latency < 2.5s for complex queries, billing < $0.001 per turn.

Execution G

- **Zero Training Skew:** `a4i_ml.poi` TVF serves as feature assem both training
- **Spatial Leaka** 5 km Geohash guarantee ge generalizatio memorized p
- **Honest Refus** The agent str queries outsi campaign foo transparent ra

Speaker Note (1:30–2:00): *"Every tier utilizes Google Cloud primitives natively: BigQuery ML trains the physical offset model; MCP Toolbox exposes parameterized SQL tools; Google ADK coordinates Gemini 2.5; and Cloud Run hosts the live experience with zero-latency response."*

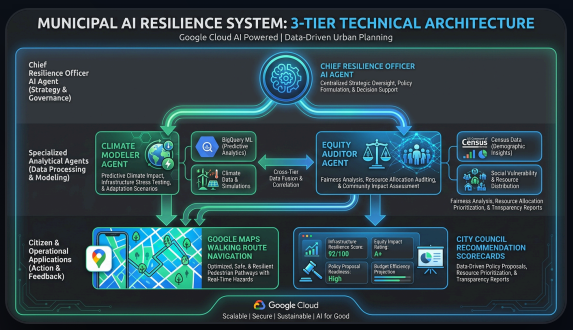
LIVE DEMONSTRATION HIGHLIGHTS · DECISION WORKFLOWS

Community Resilience & Explainable Siting

Four Core Decision Capabilities Validated on Live Atlanta Census Data

The 4 Core Decision Capabilities

- **1. Microclimate Explainability:** "Why does Site 1 run hot?" → BQML attributions isolate physical causes (+3.8°F impervious pavement, +1.2°F building density vs. -0.9°F tree canopy).
- **2. Multi-Criteria Vulnerability Ranking:** Composite score optimizes permanent sites: *Heat Offset × 7-Yr Days ≥95°F × Non-Car Households & Seniors (800m walk catchment)* with 2 km non-overlap constraint.
- **3. Post-Hoc Demographic Equity Audit:** Proves algorithmic justice. Recommended tracts serve **3.04x higher poverty rate** and **7.38x higher transit dependency** than the Atlanta baseline.
- **4. Council-Ready Policy Memo:** Auto-generates structured memos complete with GIS coordinates, annual operating estimates, displacement rationale ("Why not Site 4?"), and human sign-off.



> **Speaker Note (2:00–2:30):** *"When we ask for 3 sites, the agent ranks candidates, computes catchment demographics via the MCP Toolbox, audits equity against the city baseline,*

HACKATHON SUBMISSION & HONEST ENGINEERING

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Responsible AI, Impact & Official Submission

Everything Required to Verify, Replicate, and Evaluate the Solution

Honest Engineering Limitations & Ethical Guardrails

- **Data Reality:** Heat Watch mobile traverses capture a single hot summer afternoon; weather stations capture regional trends rather than block-level continuous temporal variance.
- **Simulation Boundary:** The model diagnoses current microclimate fever; it does not simulate post-intervention cooling deltas (cannot predict cooling from newly planted trees).
- **Distance Metric:** Catchments use 800m straight-line radii rather than street-network walking isochrones (future unlock: Google Maps Routing API).
- **Ethical Guardrail:** Race is strictly excluded from model features; it appears solely as a post-hoc verification metric to audit equitable outcomes.

Explore Qwiklabs Environment

`project=qwiklabs-gcp-04-ec3596addc43`

US Multi-Region BigQuery datasets (a4i_core, a4i_semantic, a4i_ml) & Cloud Run services.

GitHub Repository

`github.com/chintamanenirc/A4I2026-challenge-1-walkability`

Complete codebase: BigQuery SQL, BQML training, ADK agent, MCP Toolbox, and Next.js UI.

Live Demo (Cloud Run UI)

`urban-heat-ui-1042443035046.us-central1.run.app`

Interactive Next.js 16 dashboard with live Voronoi map, BQML explainability & equity audit.

Recorded Pitch Video (< 3 Minutes)

`Duration: 2m 45s (Script in FINAL_PRESENTATION_DECK.md)`

Comprehensive pitch video walkthrough covering problem, team, architecture, and live demo.

Speaker Note (2:30–3:00): *"We don't claim to have solved urban heat. We built an honest, defensible tool that helps city planners put real cooling centers where heat hurts the most vulnerable residents. Thank you."*